Reproducible research

Dr Metodi Siromahov

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Outline

- The reproducibility crisis in psychology
- Calculating the predictive value of research
- Threats to reproducibility
- What can we do to fix science?

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Reproducibility crisis

- This phrase was coined in the early 2000s to describe growing concern that many effects in the published literature were not real
- This was partly because Professor Daryl Bem of Cornell University published a study suggesting that people could see into the future….

Essay

Why Most Published Research Findings Are False

John P. A. Ioannidis

2005. PLoS Medicine, 2(8), e124. doi: 10.1371/journal.pmed.0020124

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Social priming and nudges

Daniel Kahneman's "Thinking: Fast and Slow" (2011)

- Reviews lots of research on biases in intuitive (automatic) thinking
- Many studies based on social priming
  - Subtle cues can have profound effects on behaviour
  - Write about unethical deeds -> need to wash hands
  - Read words about old age -> walk more slowly

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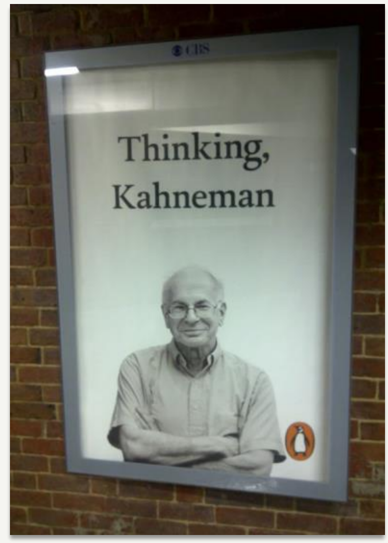
Social priming and nudges

"Disbelief is not an option.

The results are not made up, nor are they statistical flukes.

You have no choice but to accept that the major conclusions of these studies are true.

More important, you must accept that they are true about *you*."



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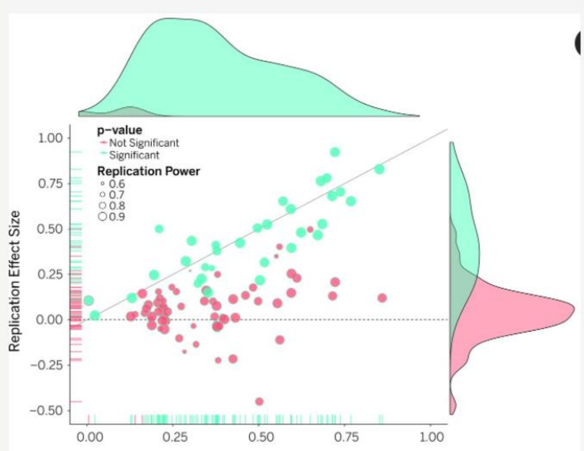
The reproducibility crisis

- Large team attempted to replicate 100 studies in top psychology journals
- 36% of replications had significant results (p<.05) vs. 97% of original studies
- 47% of original effect sizes were in the 95% confidence interval of the replication effect size;
- 39% of effects were subjectively rated to have replicated the original result

https://www.science.org/doi/10.1126/science.aac4716



Original study effect size versus replication effect size (correlation coefficients). Diagonal line represents replication effect size equal to original effect size. Dotted line represents replication effect size of 0. Points below the dotted line were effects in the opposite direction of the original. Density plots are separated by significant (blue) and nonsignificant (red) effects.

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Social priming and nudges

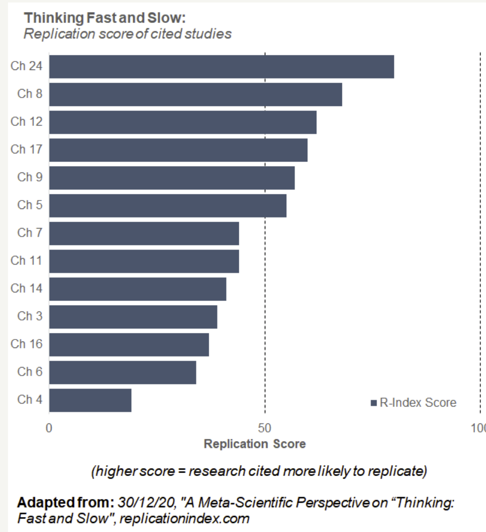
"Thinking: Fast and Slow" ten years on…

- Only 12-46% of studies are replicable
- Critique
- Kahneman (2012):

  "your field is now the poster child for doubts about the integrity of psychological research"

  "The experimental evidence for the ideas I presented in that chapter was significantly weaker than I believed when I wrote it. …

  I knew all I needed to know to moderate my enthusiasm … but I did not think it through."



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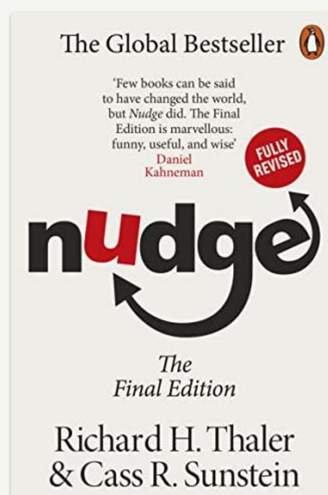
Social priming and nudges

- "Nudge" was another best-seller on how subtle cues can nudge people to change their behaviour
  - Huge implications for libertarian public policy; many governments set up Nudge Units

Part 1 - Part 2



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Which of these are real effects?

- Tests of implicit racial bias
- Power poses
- Facial feedback: forcing yourself to smile makes you feel happier
- Ego depletion: willpower is a finite resource

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Gervais & Norenzayan (2012):

- Exposing people to subtle cues related to analytical thinking decreases their religious belief)
  - (manipulation: images of a person thinking; a hard to read text font)

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NBC NEWS WATCH LIVE

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SCIENCE NEWS

Analytic Thinking Can Promote Atheism

Deliberate analytical thinking can cause people to believe less in God, according to a new study.

April 26, 2012, 7:22 PM BST / Source: LiveScience

By Stephanie Pappas, LiveScience Senior Writer

Deliberate analytical thinking can cause people to believe less in God, according to a new study.

The researchers, who found that religious belief arises from gut feelings, were quick to say their study was not a referendum on the value of religion. Both analytical thinking and the intuitive processing that seems to promote religious beliefs are important, said study researcher Will Gervais.

PLoS One. 2017; 12(2): e0172636.
Published online 2017 Feb 24. doi: 10.1371/journal.pone.0172636
PMCID: PMC5325262
PMID: 28234942

Direct replication of Gervais & Norenzayan (2012): No evidence that analytic thinking decreases religious belief

Clinton Sanchez,[1,2] Brian Sundermeier,[3] Kenneth Gray,[4] and Robert J. Calin-Jageman[1,*]

Michiel van Elk, Editor

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Associated Data

▸ Supplementary Materials

▸ Data Availability Statement

Abstract Go to: ▸

Gervais & Norenzayan (2012) reported in *Science* a series of 4 experiments in which manipulations intended to foster analytic thinking decreased religious belief. We conducted a precise, large, multi-site pre-registered replication of one of these experiments. We observed little to no effect of the experimental manipulation on religious belief (d = 0.07 in the wrong direction, 95% CI[-0.12, 0.25], N = 941). The original finding does not seem to provide reliable or valid evidence that analytic thinking causes a decrease in religious belief.

Author's analysis and response:
http://willgervais.com/blog/2017/3/2/post-publication-peer-review

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Ice Cube - Check Yo Self (Remix) (Official Music Video)

"Chickity-check yo' self before you wreck yo' self"
- Ice Cube

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It's not just social psychology!

- Medicine (e.g., cancer research)
- Neuroscience (esp. clinical and computational)
- Brain imaging
- Pharmacological research

Believe it or not: how much can we rely on published data on potential drug targets?

Our findings are mirrored by 'gut feelings' expressed in personal communications with scientists from academia or other companies, as well as published observations. An unspoken rule among early-stage venture capital firms that "at least 50% of published studies, even those in top-tier academic journals, can't be repeated with the same conclusions by an industrial lab" has been recently reported (see Further information) and discussed[4]. The

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Positive predictive value

What is the probability that a significant research finding you read about in the literature is actually true?

$$PPV = \frac{p(true\ positive\ result)}{p(true\ positive\ result) + p(false\ positive\ result)}$$

$$p(true\ positive\ result) = p(hIsTrue) * (1 - \beta)$$

$$p(false\ positive\ result) = \big(1 - p(hIsTrue)\big) * \alpha$$

Ioannidis, J. P. (2005). Why most published research findings are false. *PLoS medicine*, *2*(8), e124.

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Positive predictive value

What is the probability that a significant research finding you read about in the literature is actually true?

$$PPV = \frac{p(true\ positive\ result)}{p(true\ positive\ result) + p(false\ positive\ result)}$$

Probability that the hypothesis is true (before study)

Statistical Power

$$= \frac{p(hIsTrue) * (1 - \beta)}{p(hIsTrue) * (1 - \beta) + \left(1 - p(hIsTrue)\right) * \alpha}$$

Alpha level

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Positive predictive value

What is the probability that a significant research finding you read about in the literature is actually true?

Statistical Power
assumed to be .8

$$PPV = \frac{p(hIsTrue) * .80}{p(hIsTrue) * .80 + \left(1 - p(hIsTrue)\right) * .05}$$

Alpha level
Typically set
to .05

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Positive predictive value

If probability that H = True is 80%, our PPV is .98. This means that out of 100 studies in the literature that find a positive result, 2 will be false.

$$PPV = \frac{.80 * .80}{.80 * .80 + (1 - .80) * .05} = .98$$

However, 80% is probably an overestimate. Remember: if we are certain a hypothesis is true, then there would be no point in doing a study..

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Positive predictive value

- In **highly novel** areas it would not be unreasonable to estimate that only 10% of hypotheses are true
- If probability that the hypothesis is true is only 10%, then our PPV is .64. This means out of 100 studies in the literature that find a positive result, 36 will be false.

$$PPV = \frac{.10 * .80}{.10 * .80 + (1 - .10) * 0.05} = .64$$

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But most studies are underpowered

- Empirical estimates of the power of studies in neuroscience suggest median power of published studies is 8%-31% (Button et al, 2013)
- If probability that the hypothesis is true is 10% and power is 20%, then our PPV is .31.

➔ out of 100 studies in the literature that find a positive result, 69 will be false.

$$PPV = \frac{.10 * .20}{.10 * 0.2 + (1 - .10) * 0.05} = .31$$

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Threats to reproducibility

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The obvious problems

- Fraud
- Sloppiness
  - https://forbetterscience.com/2024/01/02/dana-farberications-at-harvard-university/
  - https://www.nytimes.com/interactive/2022/10/29/opinion/science-fraud-image-manipulation-photoshop.html

The New York Times

OPINION

Science Has a Nasty Photoshopping Problem

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UK ▸ UK politics Education Media Society Law Scotland Wales

Science

'A lot of it is sloppiness': the biologist who finds flaws in scientific papers

Sholto David has flagged thousands of papers, most because of concerns over potential image manipulation

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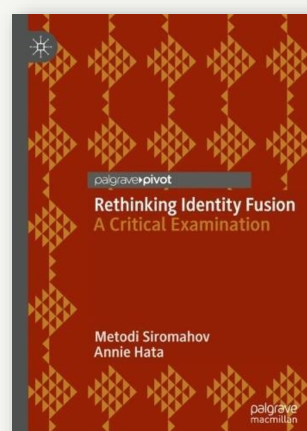
More intractable problems

- Poorly defined constructs
- Poorly operationalised constructs
- Huge variability in how the same data is analysed



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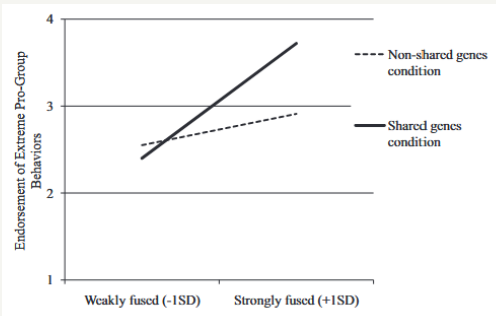
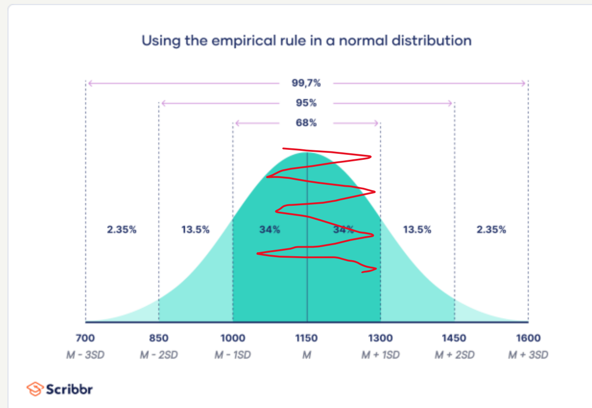


Figure 2. Study 2: Fusion and genetic race prime interactively predict endorsement of extreme pro-group behaviors in China. Values for weakly fused and strongly fused represent ± 1 *SD* from the mean ($M = 3.86$, $SD = 0.84$).



Swann Jr, W. B., Buhrmester, M. D., Gómez, A., Jetten, J., Bastian, B., Vázquez, A., ... & Zhang, A. (2014). What makes a group worth dying for? Identity fusion fosters perception of familial ties, promoting self-sacrifice. *Journal of personality and social psychology*, *106*(6), 912.

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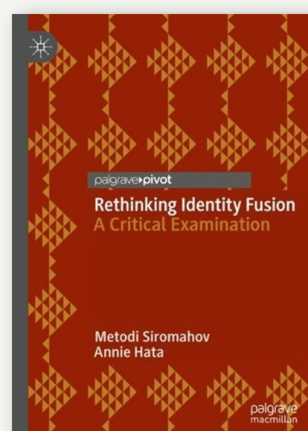
More intractable problems

- Poorly defined constructs
- Poorly operationalised constructs
- Huge variability in how the same data is analysed
- Authors misinterpreting or outright not reading earlier papers



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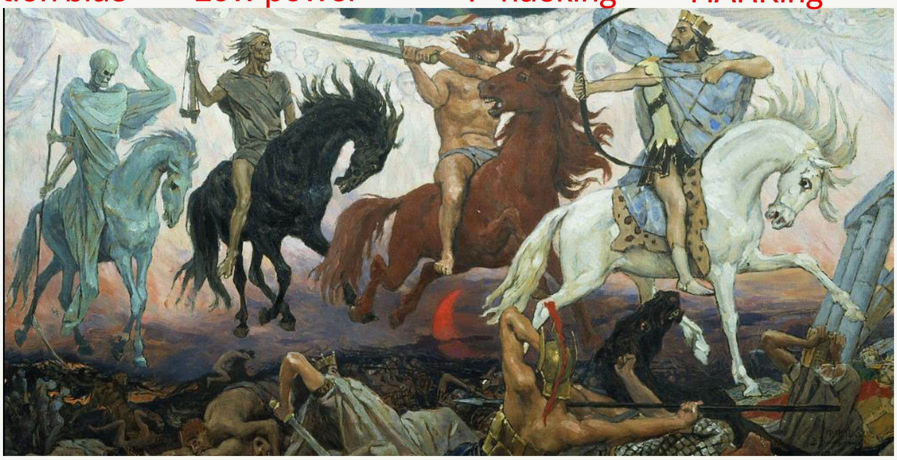
Other threats to the scientific process

Publication bias Low power P-hacking HARKing



Bishop (2019)
Nature **568**, 435

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Publication bias

- Not many studies with null results get published – instead, they end up in researcher's "file drawer".
- This is a problem, because given the typical .05 significance level, 1 in 20 studies looking for an effect that does not exist will find a significant p value.
- That one study is more likely to end up in the literature and be read, leading people to conclude that the effect does exist.

File drawer problem

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Low power

- Power depends on **sample size** and **effect size**
  - Many studies in psychology and neuroscience have small samples
  - Most effects we look for are small because we are looking at complex phenomenon

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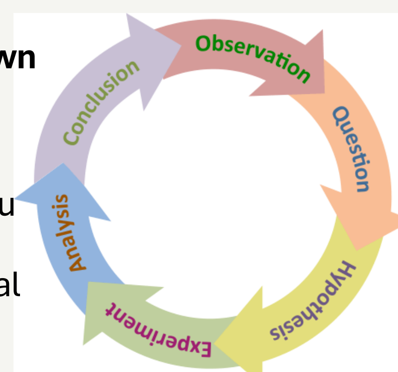
HARKing

- **HARKing: Hypothesising after the results are known**
- Hypotheses should be set in advance, based on the literature, before you collect data.
- If you find something other than what you expect, you may feel like you should change your hypotheses so that you get the "right" answer.. but this causes a fatal flaw in the scientific process.



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Do not do this!

Explicitly advises HARKing!

Writing the Empirical Journal Article
Daryl J. Bem
The Compleat Academic: A Practical Guide for the Beginning Social Scientist, 2nd Edition. Washington, DC: American *Psychological* Association, 2004.

Which Article Should You Write?
There are two possible articles you can write: (a) the article you planned to write when you designed your study or (b) the article that makes the most sense now that you have seen the results. They are rarely the same, and the correct answer is (b).

"This book provides invaluable guidance that will help new academics plan, play, and ultimately win the academic career game."

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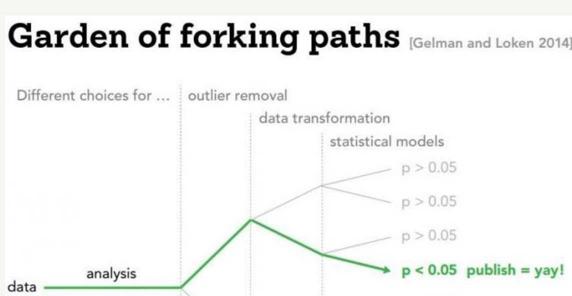
p-hacking

- We have to make lots of decisions when we analyse data
- If we try all the different possibilities and only report the one that gives us the "right" answer ($p<.05$) we increase the chances of reporting a false finding



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Do not do this!

Explicitly advises p hacking!

Writing the Empirical Journal Article
Daryl J. Bem
The Compleat Academic: A Practical Guide for the Beginning Social Scientist, 2nd Edition. Washington, DC: American *Psychological* Association, 2004.

re Data Analysis: Examine them from every angle. Analyze the sexes separately. Make up new composite indexes. If a datum suggests a new hypothesis, try to find additional evidence for it elsewhere in the data. If you see dim traces of interesting patterns, try to reorganize the data to bring them into bolder relief. If there are participants you don't like, or trials, observers, or interviewers who gave you anomalous results, drop them (temporarily). Go on a fishing expedition for something— anything —interesting.

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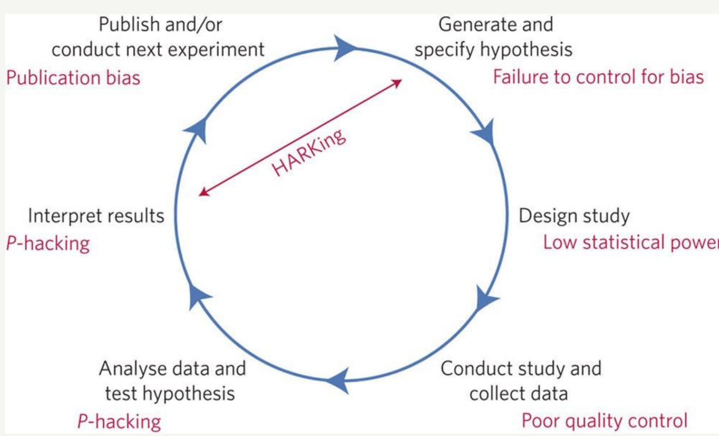
Summary



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Open science as a solution

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Science has an incentive problem

What's best for science

Transparent and high-quality research, regardless of outcome

What's best for scientists

Producing a lot of "good results"

see Nosek, Spies & Motyl (2012). *Perspectives on Psychological Science, 7(6)*: 615–631

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Open science

- The Open Science movement has grown out of concern about reproducibility
- It aims to change the incentive structures in science and increase transparency

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Preregistration

- Researchers submit a detailed plan of their study including hypotheses, methods and analysis decisions on a platform
- Reduces the possibility of intention or unintentional HARKing and p-hacking
- Increases the publication of null findings

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Open data

- Share your data!
- We have used lots of open data sets in practicals on this course
- It allows others to directly reproduce analysis which helps identify errors
- It also means data can be used for other purposes to answer new research questions (secondary data analysis), which makes the most of existing data.

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Reproducible analysis

- Conducting data analysis using a script-based program like R increases computation reproducibility:
- It is a detailed record of your analytical decisions
- It allows scientists to easily reproduce your results with the same data
- It also makes replication much easier as the code can be applied to a new dataset from a different study.

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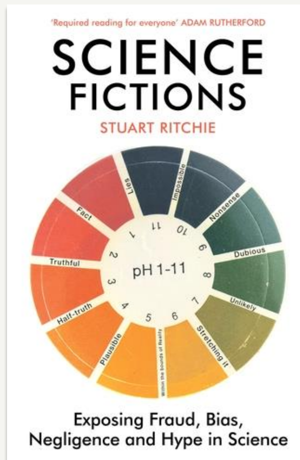
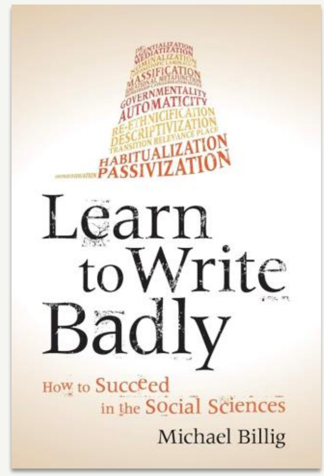
Further reading

Retraction Watch

Tracking retractions as a window into the scientific process

https://retractionwatch.com/





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Thank you

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